SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.3: Vectors I — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.3: Vectors I
Driving Question	DRIVING QUESTION: Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is a Vector? Quantities That Have Direction**	Students observed that some quantities — such as the ferry's push across Lake Victoria — cannot be fully described by a number alone. Using a class activity with arrows drawn on grid paper, students compared a scalar (e.g., the temperature in Kisumu: 28 °C) with a vector (e.g., the wind blowing 15 km/h due south). They noted that the arrow's length represented magnitude and its head represented direction. The anchoring phenomenon was introduced: the Kisumu ferry departed on a straight heading toward Mbita but arrived off-course because the lake current pushed it sideways.	Students learned that a vector is a quantity that has both magnitude (size) and direction, and that scalars have magnitude only. They identified real-world Kenyan examples: a matatu's speed is a scalar, but its velocity (speed + direction along Nairobi–Nakuru highway) is a vector. They also learned that vectors are represented as directed line segments (arrows), that equal vectors have the same magnitude and direction regardless of starting point, and that a negative vector reverses direction while keeping the same magnitude.	Teacher should emphasise that the ferry problem cannot be solved with arithmetic alone — the lake current and the engine thrust are both vectors, so the ferry's true path is the combined effect of two directed quantities. Cold-call students to distinguish scalars from vectors using at least three local examples (e.g., mass of a sack of maize vs. force applied to push it; distance walked vs. displacement from Kisumu Bus Station to Jomo Kenyatta Sports Ground). Reinforce that direction is non-negotiable: ignoring it is precisely why the ferry missed Mbita. Provide 5–8 seconds of wait time before accepting answers. Pedagogical note: use a physical demonstration — have a student walk three steps east, then the wind (teacher) nudges them two steps south; ask the class where they ended up. This embeds the phenomenon kinesthetically.
Lesson 2 **Representing Vectors — Column Vectors and Position Vectors**	Students observed how the ferry's engine thrust and the lake current, previously described only in words and arrows, could be written as precise column vectors on a	Students learned that a column vector (x / y) uses a top entry for horizontal displacement and a bottom entry for vertical displacement, with sign indicating direction	Teacher should display a large grid map of Lake Victoria on the board and ask students: 'If the engine pushes the ferry 8 km due east, what is the column vector?'

	Cartesian grid overlaid on a simplified map of Lake Victoria. For example, the engine thrust was represented as (8 / 0) (8 km east, 0 km north) and the lake current as (0 / −3) (0 km east, 3 km south). Students also plotted position vectors from the origin (Kisumu Port) to several landmark points — Mbita, Rusinga Island, and Homa Bay — reading off column vectors from their coordinates.	(positive = right/up; negative = left/down). They learned that a position vector describes the location of a point relative to a fixed origin. They practised converting arrow diagrams to column vectors and vice versa, and they identified that two vectors are equal if and only if their column vectors are identical — confirming that equal vectors need not start at the same point.	(Answer: (8 / 0).) Then ask: 'The current pushes it 3 km due south — what is that vector?' (Answer: (0 / −3).) Cold-call at least four students to write column vectors for different journeys on the grid. Pedagogical note: stress sign conventions rigorously here — a single sign error is enough to send the ferry in the wrong direction. Reinforce that the column vector is not a coordinate pair but a displacement instruction. Wait time of 6 seconds is recommended before cold-calling on sign questions. Connect back to the DQ: the captain needs both of these column vectors before any calculation can begin.
Lesson 3 **Vector Addition — Triangle and Parallelogram Rules**	Students observed two physical demonstrations: (1) a tip-to-tail walk on the classroom floor where one student walked the engine-thrust vector (8 steps east) and then a second student continued from that endpoint with the current vector (3 steps south), arriving at the resultant endpoint — illustrating the triangle law; (2) on grid paper, students drew both vectors from the same starting point and completed the parallelogram, confirming that the diagonal gave the same resultant. Both methods produced the resultant vector (8 / −3) for the ferry scenario.	Students learned that the triangle law states: place vectors tip-to-tail; the resultant is the vector from the free tail to the free tip. The parallelogram law states: place both vectors tail-to-tail; the resultant is the diagonal of the completed parallelogram. Both laws always give the same resultant. Students practised adding column vectors algebraically — adding corresponding components: (a / b) + (c / d) = (a+c / b+d) — and verified graphically. They also learned that vector addition is commutative (a + b = b + a) and associative.	Teacher should ask: 'Using the triangle law, where does the ferry actually end up if the engine pushes (8 / 0) and the current pulls (0 / −3)?' Guide students to compute (8 / 0) + (0 / −3) = (8 / −3). Cold-call three students to add different vector pairs representing other Lake Victoria journeys (e.g., supply boat from Kisumu to Homa Bay with a northward current). Pedagogical note: many students confuse the resultant direction (the diagonal) with one of the component vectors — use a contrasting colour for the resultant arrow and emphasise it is a new vector, not merely a longer version of one component. Wait 7 seconds after posing the parallelogram question before accepting responses. Tie back to DQ: the captain must add the engine vector and current vector to predict the true resultant path.
Lesson 4 **Vector Subtraction and the Negative Vector**	Students observed that subtracting a vector is equivalent to adding its negative (reverse). Using the ferry scenario, they explored: 'If the ferry arrives at point R = (8 / −3) but wanted to reach Mbita at M = (8 / 0), what correction vector is needed?' They plotted both endpoints on a grid, drew the arrow from R to M, and read off the correction vector (0 / 3) — a northward push equal and opposite to the current. They verified that M − R = (8 / 0) − (8 / −3) = (0 / 3).	Students learned that the negative of a vector reverses its direction while keeping its magnitude: if a = (x / y) then −a = (−x / −y). Vector subtraction a − b is defined as a + (−b). They applied subtraction to find unknown vectors in journeys: given two position vectors, the displacement vector between them equals the destination position vector minus the start position vector. This was applied to multi-stop routes on Lake Victoria (e.g., Kisumu → Kendu Bay → Mbita) to find missing legs of the	Teacher should write on the board: 'The ferry's engine produces vector e. The current produces vector c. The ferry wants to end up at Mbita. What vector must the captain aim for?' Lead students to see that the required engine vector equals the desired resultant minus the current: e = resultant − c. Cold-call five students to solve subtraction problems involving position vectors of named Kenyan lake ports. Pedagogical note: insist on algebraic working alongside the diagram — students

		journey.	who draw without algebra often make sign errors. Emphasise that the negative vector is geometrically the same arrow flipped 180°. Wait time: 8 seconds for subtraction problems. Connect to DQ: the captain can now calculate the exact correction needed to counteract any current.
Lesson 5 **Scalar Multiplication of Vectors and Parallel Vectors**	Students observed the effect of multiplying the ferry's engine vector by different scalars. Starting with the engine vector e = (4 / 0), they computed 2e = (8 / 0), 3e = (12 / 0), and −1e = (−4 / 0), then drew all results on grid paper. They noticed that positive scalar multiples stretched the vector along the same line (eastward) while negative scalars reversed it (westward). Students also compared e = (2 / 3) and f = (6 / 9) and discovered that f = 3e — both vectors point in the same direction, confirming they are parallel.	Multiplying a vector by a positive scalar scales its magnitude while preserving its direction; multiplying by a negative scalar reverses the direction. For vector a = (x / y) and scalar k: ka = (kx / ky). Two vectors are parallel if and only if one is a scalar multiple of the other. Parallel vectors pointing the same way have a positive scalar; anti-parallel vectors have a negative scalar. Students learned to test for parallelism algebraically and to find unknown scalars. Practical application: a ferry travelling at double speed in the same direction has a velocity vector that is a scalar multiple (k = 2) of its original velocity vector.	Teacher should pose: 'The Mbita supply boat travels with velocity vector (3 / 2). A faster cargo vessel travels with velocity vector (9 / 6). Are they moving in the same direction?' Guide students to check (9 / 6) = 3 × (3 / 2) — yes, parallel, same direction. Cold-call four students with similar parallel/non-parallel tests using Kenya-relevant contexts (e.g., two fishing boats on Lake Victoria, two lorries on the Kisumu–Nairobi highway). Pedagogical note: students often confuse 'parallel vectors' with 'equal vectors' — stress that parallel vectors share direction (or opposite direction) but need not have equal magnitude. Wait time: 6 seconds. Connect to DQ: scalar multiplication allows the captain to model what happens if the current doubles in strength — the correction vector must also scale accordingly.
Lesson 6 **Magnitude of a Vector — How Far Did the Ferry Really Travel?**	Students observed that after adding the engine vector (8 / 0) and the current vector (0 / −3) to get the resultant (8 / −3), the arrow on the grid clearly travelled a longer path than simply 8 km east. Using rulers on their grid paper, they measured the resultant arrow's length and compared it to 8 cm. They then applied the Pythagorean theorem to a right triangle formed by the horizontal component (8) and vertical component (3), computing $\sqrt{8^2 + 3^2} = \sqrt{64 + 9} = \sqrt{73} \approx 8.54$ km. The ferry actually travelled approximately 8.54 km, not 8 km.	The magnitude of a vector v = (x / y) is $\lvert v \rvert = \sqrt{x^2 + y^2}$, derived directly from the Pythagorean theorem. Magnitude is always non-negative. Students practised calculating magnitudes for a range of vectors representing Lake Victoria journeys, including cases with non-integer answers. They also learned that two vectors can have different directions but equal magnitude (e.g., (3 / 4) and (−3 / 4) both have magnitude 5). The unit of magnitude inherits the unit of the original vector (km, m/s, N, etc.).	Teacher should ask: 'The ferry captain reported travelling 8 km — but the lake current also moved the ferry 3 km south. Did the ferry actually travel 8 km, or more?' After students compute $\sqrt{73} \approx 8.54$ km, emphasise that the captain underestimated the distance because they ignored the current's contribution. Cold-call three students to calculate magnitudes for other journeys (e.g., a boat with resultant vector (5 / 12) — magnitude 13; a wind vector (−6 / 8) — magnitude 10). Pedagogical note: require students to draw the right triangle alongside every calculation — this cements the geometric meaning of the formula and prevents rote application. Wait time: 7 seconds. Connect to DQ: knowing the actual distance travelled helps the captain calculate fuel consumption and arrival time

			more accurately.
Lesson 7 **Collinear Points and Vector Paths**	Students observed a scenario in which the ferry made stops at three ports — Kisumu (K), an intermediate point P, and Mbita (M) — and were given position vectors for all three. Using these, they computed vectors KP and KM and tested whether KM = k · KP for some scalar k. In one case the points were collinear (all three on the same straight route along the lake shore); in another, P was off the direct route, producing vectors that were not scalar multiples and confirming non-collinearity. Students also traced multi-leg vector paths (K → P → M) and expressed them as a single vector sum.	Students learned that three points A, B, C are collinear if and only if vector AB and vector AC are parallel (i.e., one is a scalar multiple of the other). The test: express both vectors as column vectors from a common point and check whether AB = k · AC. They also practised constructing vector paths: the vector from A to C via B equals vector AB + vector BC (tip-to-tail). Students applied this to trace ferry routes with multiple stops and to verify whether three GPS waypoints on Lake Victoria lie on a straight course.	Teacher should display three points on a Lake Victoria map grid and ask: 'Can the ferry travel in a perfectly straight line through all three ports, or must it turn?' Students compute the relevant vectors and test the scalar multiple condition. Cold-call five students — vary cases between collinear and non-collinear to keep all students alert. Pedagogical note: a common error is testing AB and BC (consecutive vectors) instead of AB and AC (vectors from a common point) — explicitly demonstrate why the common-point test is the correct method. Provide the counter-example on the board. Wait time: 8 seconds before cold-calling on the scalar-multiple test. Connect to DQ: a captain planning a multi-stop route across Lake Victoria must know whether waypoints are collinear to determine if turning is required.
Lesson 8 **Translation Vectors and Unit Synthesis — Solving the Ferry Problem**	Students observed a full synthesis scenario: the ferry captain knows (1) the desired destination vector to Mbita from Kisumu (the intended resultant), (2) the lake current vector (a known external force), and must calculate (3) the required engine heading vector. Students drew all three vectors on a Lake Victoria grid, identified the vector equation: engine vector + current vector = resultant vector, and rearranged to solve for the engine vector. They also explored translation vectors — how a shape (e.g., a fishing zone boundary on the lake) is moved by adding a fixed translation vector to each vertex's position vector — connecting vectors to coordinate geometry.	Students learned that: (1) two simultaneous vectors — the engine's push and the current's pull — combine by addition into a single resultant, and the engine heading can be found by subtracting the current vector from the desired resultant; (2) a translation vector moves every point in a plane by the same displacement, preserving shape and size; (3) the full toolkit of Vectors I (representation, addition, subtraction, scalar multiplication, magnitude, collinearity, and translation) provides the captain — and any navigator — with complete mathematical control over direction and displacement. Students revisited the driving question and articulated a full answer using precise vector language.	Teacher should facilitate a whole-class solution to the anchoring problem: 'The captain wants to reach Mbita — displacement vector (8 / 0). The current will push the ferry (0 / −3). What heading must the engine produce?' Students solve: engine vector = (8 / 0) − (0 / −3) = (8 / 3), magnitude = $\sqrt{(64 + 9)} = \sqrt{73} \approx 8.54$ km. The captain must aim slightly north of east. Cold-call six students in sequence to narrate each step of the solution. Pedagogical note: this lesson is the payoff for all prior lessons — insist that students explicitly name which concept (addition, subtraction, magnitude, etc.) they are using at each step, reinforcing the coherence of the unit. Ask: 'Which lesson gave us this tool?' for each step. Wait time: 10 seconds for the synthesis question. Close by returning to the DQ on the board and inviting students to answer it in their own words — a formative assessment of the full unit.